CARMA – Document Requirements Product

Document Version: v1.0

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2. Technical Background

The CARMA system is a mobile platform, which monitors and grades driving behavior in real time.

The system uses the IMU sensors (Accelerometer & Gyroscope) and the mobile device's GPS sensor to detect safety events such as sudden braking, sharp turns, and distractions (using a phone).

The app samples the travel data, performs initial processing, and sends commands to the server for the purpose of calculating a CARMA score, managing a gamification system (CARMA points), and operating a marketplace to realize benefits in businesses.

At the end of the trip, an alert will be sent that includes a summary and personal insights, in order to give positive reinforcement to the driver or encourage improvement in his performance.

3. User classification

3.1. Driver: An end user whose driving is monitored, earning points and redeeming vouchers.

3.2. Business: A user with permissions to manage a benefits catalog and validate purchase vouchers.

3.3. Administrator: Global parameters management of the ranking algorithm, database management, and technical support.

4. Functional Requirements

4.1. Platform Requirements

4.1.1. The app will support Android (version 11 or later) and iOS (version 14 or later) operating systems.

4.1.2. The system will support linking up to 2 vehicle devices (Bluetooth IDs) to each user profile.

4.2. User Identification Requirements

4.2.1. User authentication will be done during installation, using a phone number and a one-time code (OTP) sent via SMS.

4.2.2. When registering for the app, the user will fill in details such as name, age, and a photo of a driver's license thatverifies his identity.

4.3. UI/UX Requirements

4.3.1. Autoplay

4.3.1.1. The app will run a Foreground Service and start sampling data after detecting Bluetooth pairing with the vehicle's system.

4.3.2. Interface Language

4.3.2.1. The system will support Hebrew and English.

4.3.3. User feedback

4.3.3.1. After each unusual event during the ride (braking/turning), a short visual indicator (if the screen is on) and sound (can be adjusted in the settings) will be displayed.

4.3.4. Show map

4.3.4.1. Catalog the benefits as a list or on top of the Google Maps API with the business location pins.

4.3.5. Scoring and gamification

4.3.5.1. At the end of a trip, the user will receive a score for his trip, "CARMA Score", which is calculated by the system's algorithm (Appendix C).

4.3.5.2. The system will update the user's "CARMA Points" balance at the end of the trip.

4.3.5.3. The system will manage a "roadmap" of 10 user levels, with the transition from one level to another conditional on cumulative points accumulation (Appendix D).

4.3.6. Marketplace & Redemption

4.3.6.1. The app will display a location-based benefits catalog within a radius of the user's choice.

4.3.6.2. When redeeming a benefit , a one-time QR code will be generated for display to the business.

4.3.6.3. Efficiency - Redeem a Marketplace benefit in a maximum of 3 clicks.

4.4. Internal logic

4.4.1. Sensor Monitoring: The system will sample the IMU sensors at a frequency of at least 10Hz.

4.4.2. Event detection algorithm: The system will detect anomalous events listed in sub-Appendix C1.

4.4.3. Calculating a trip score: The score will be calculated at the end of each trip according to a defined formula (Appendix C).

4.4.4. Aggregation management: CARMA Points will be added to the user's pool only after successful synchronization of the payload with the server.

4.4.5. Prerequisites for calculating a grade: (Appendix E).

4.5. Accessibility

4.5.1. Color contrast: WCAG 2.1AA compliant.

4.5.2. TalkBack (Android/VoiceOver (IOS) support.

4.5.3. Minimum font size: 14pt.

5. Technical requirements

5.1. Performance

5.1.1. Dashboard loading time should not exceed 3 seconds on a G4 connection.

5.1.2. Active driving battery consumption should not exceed 5% per hour (without charging connection).

5.1.3. The system will support 2,000 simultaneous active users (in this version) without compromising server response times.

5.1.4. The size of the app should not exceed 80MB.

5.1.5. Loading time for the "Store" page – max 2 seconds on 4G.

5.1.6. Time to receive a ride score – max 5 seconds on 4G.

5.1.7. IMU sensor sampling frequency – see 4.4.1.

5.1.8. In the event of a network disconnection while traveling, the app will store the sensor data locally (Local Cache) and synchronize with the server as the connection resumes.

5.2. Security

5.2.1. All communication between the application and the server will be conducted on the HTTPS/TLS 1.3 protocol.

5.2.2. The user's location data and phone number wereencrypted in the database after the trip was processed.

5.2.3. The system will comply with GDPR (European Union Privacy Protection Regulation) standards, including the option to self-delete the account and all information linked to it through the settings menu.

5.2.4. Limiting login attempts – 5 incorrect attempts before a temporary lockdown of 15 minutes.

5.2.5. A QR code to redeem a benefit will be valid for 5 minutes only.

5.3. Database Structure

5.3.1. Data Entities

5.3.1.1. User: (id, name, phone, points, level, license_img_url, language, created_at, last_logged)

5.3.1.2. Trip: (id, user-id, start_time, end_time, avg_score, distance, events_array)

5.3.1.3. Reward: (id, business-id, description, cost_points, expiry_date)

5.3.1.4. Businesses: (id, name, category, location_lat, location_lng)

5.3.1.5. Redemption: (id, user-id, reward-id, qr_code, status, created_at, expired_at)

5.3.1.6. Event: (id, trip-id, type, severity, timestamp, sensor_data)

5.3.1.7. Level: (id, name, min_points, discount_pct, bonus_multiplier)

5.3.2. Connections

5.3.2.1. User–Trips: One user can make many trips (one-to-many).

5.3.2.2. User-Rewards: One user can get multiple benefits (one-to-many).

5.3.2.3. User–Redemptions: A user can make multiple implementations (one-to-many).

5.3.2.4. User–Level: A Many-to-One relationship (each user belongs to one level).

5.3.2.5. Business-Rewards: One merchant can offer multiple benefits (one-to-many).

5.3.2.6. Trip–Events: One trip can include multiple events (one-to-many).

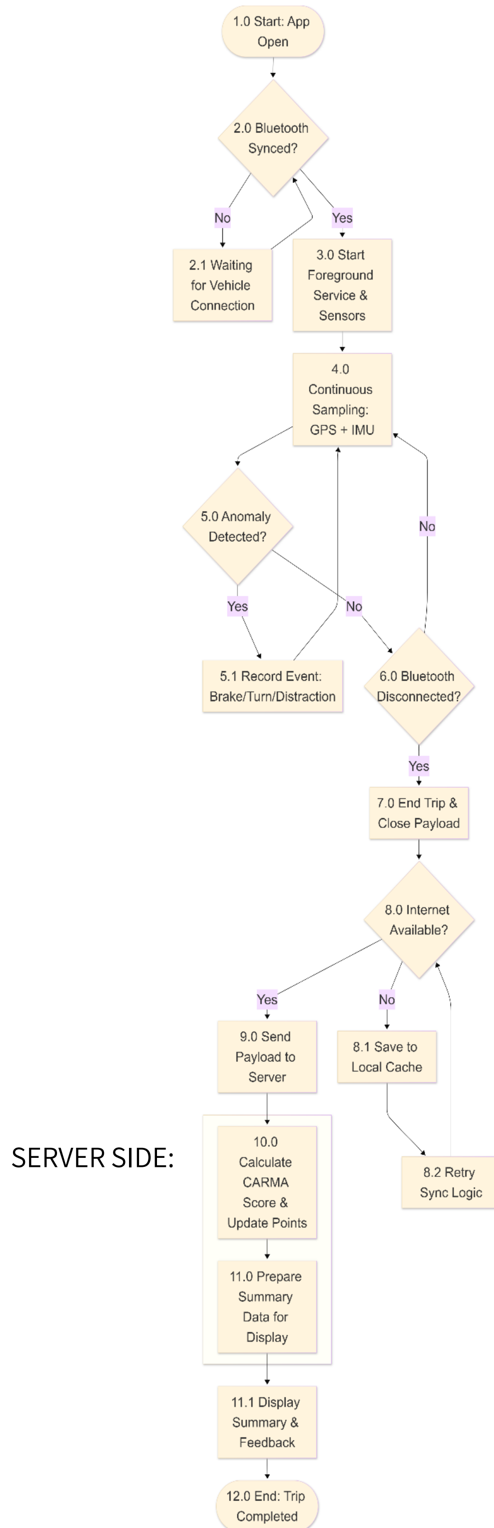
5.3.2.7. Reward-Redemptions: A benefit can be redeemed multiple times (one-to-many).

6. Prioritizing features

Explanation	Priority	Feature
System Core	Must-Have	Driving Monitoring (IMU/GPS) and Score Calculation
Basic Security	Must-Have	User identification and basic registration
A central motivational mechanism	Must-Have	Accumulation of CARMA Points and Update fromaZen
Direct Business Value	Must-Have	Marketplace and redemption code generation
Improve reliability in the field	Should-Have	Local Cache
Significant incentive for the user	Should-Have	Comparison of Community Users
Expanding your target audience	Should-Have	English Support
Boost user engagement	Nice-to-Have	Roadmap 10 Levels
Improve user experience	Nice-to-Have	Support for linking 2 vehicle devices (BT IDs)
Positive reinforcement	Nice-to-Have	Notification to the user when a trip is over
User Convenience	Nice-to-Have	Autoplay
Expanding Target Audience User Convenience	Nice-to-Have	Support for additional languages

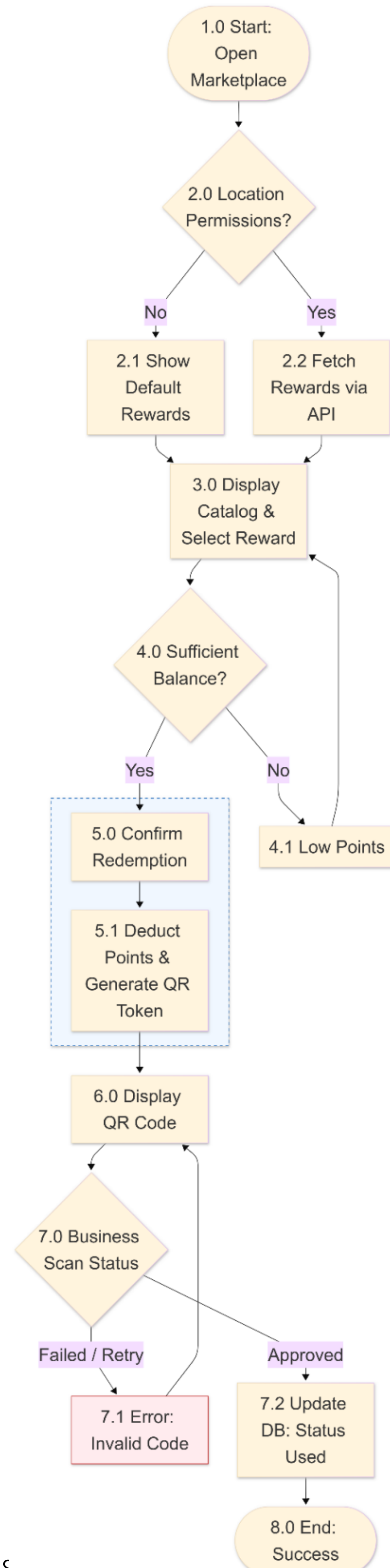
7. Wireframes &Flowcharts

Travel Diagram:

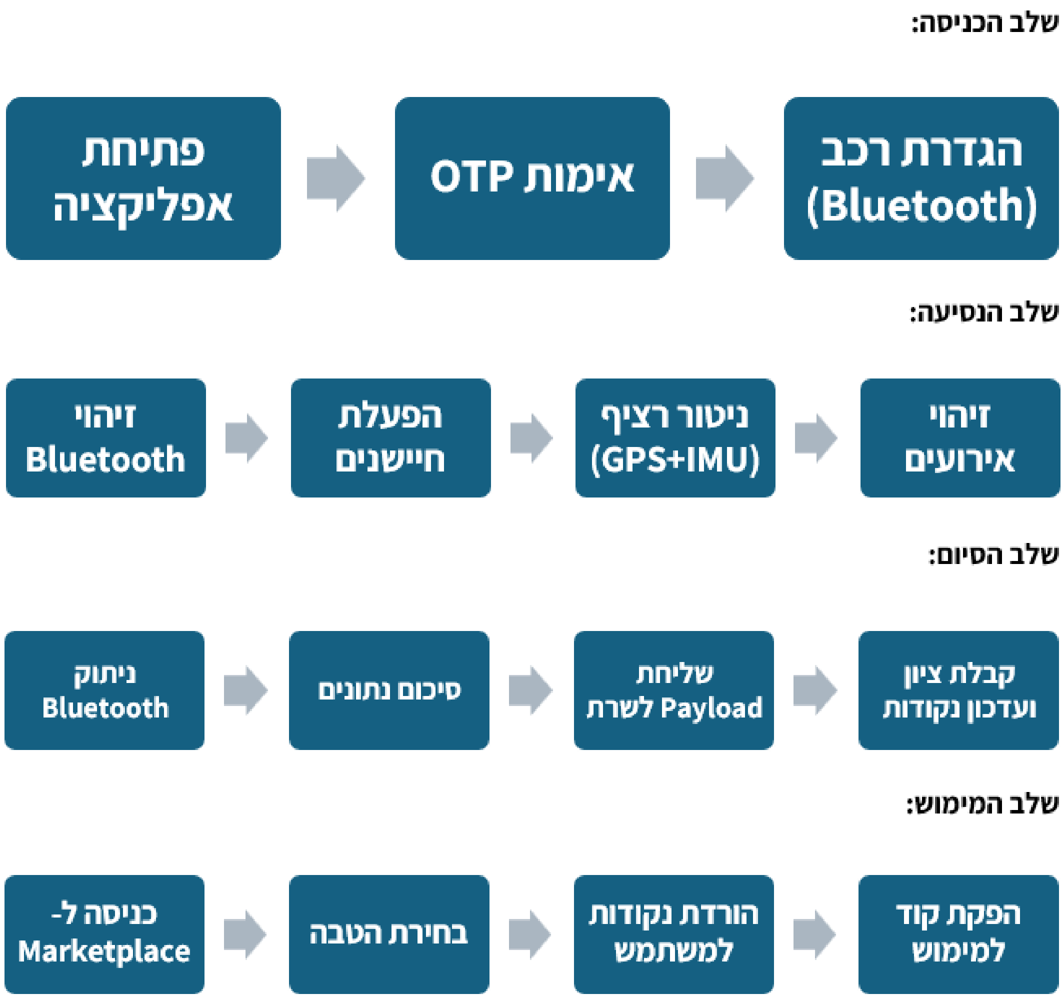


The stage of redeeming the credits:

SERVER SIDE – TRANSACTION:



Registration Step:



8. Out of Scope

8.1. Integration with external hardware components (such as OBD-II).

8.2. Redeem benefits without an active internet connection (offline redemption).

8.3. Chat or direct communication between drivers.

8.4. Integration with other navigation systems.

8.5. Support multiple drivers on the same vehicle.

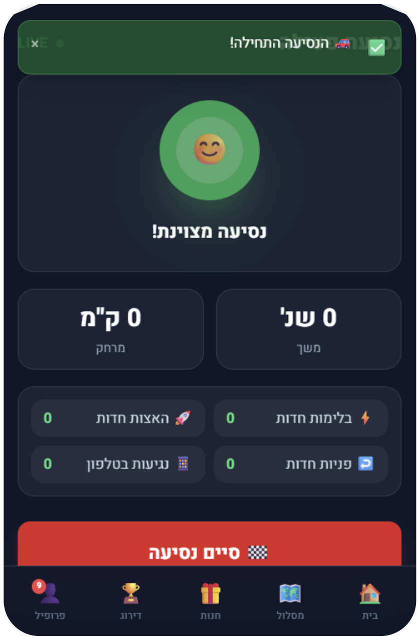
8.6. Share travel scores on social media.

8.7. A management interface for businesses via the web (not just through the app).

Appendices

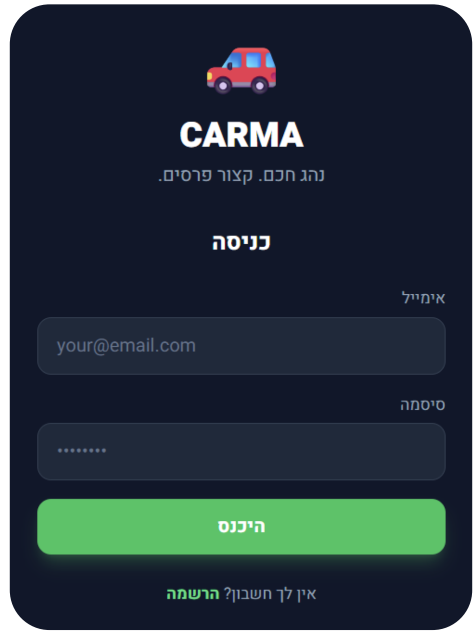
Appendix A - App Screens

Beginning of the journey:

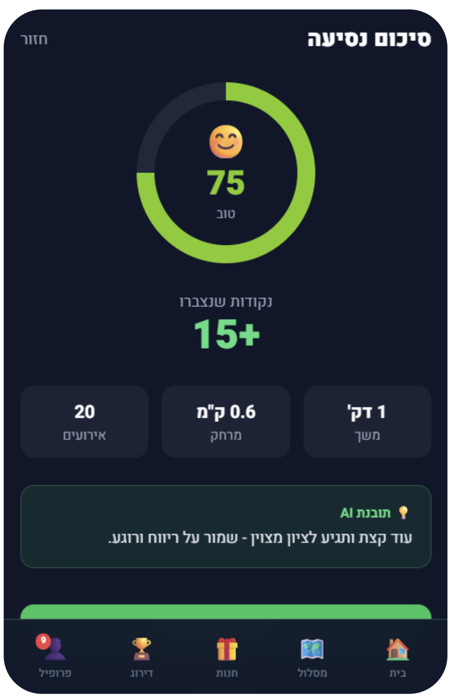


Home Screen:

Login:



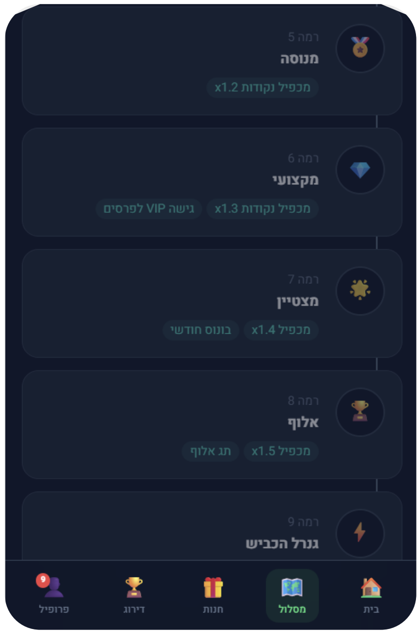
Travel summary:



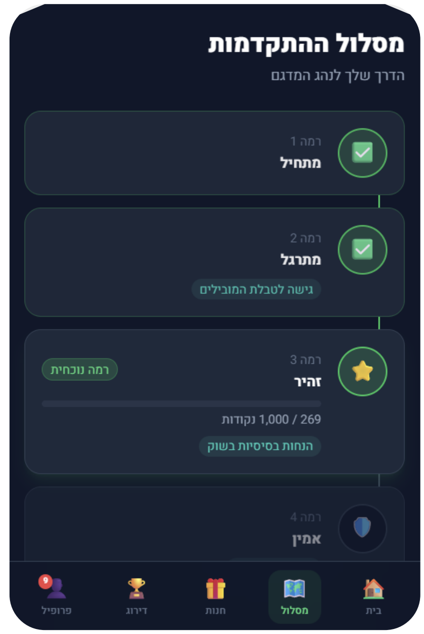
Dangerous Travel:

Identifying risky behavior:

Follow-up track:

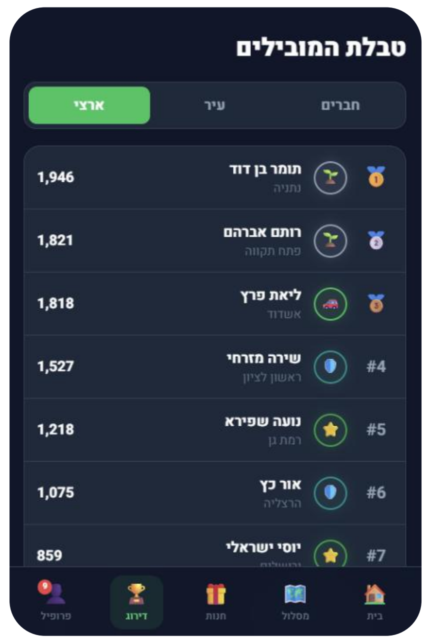


Rank Progression Track:

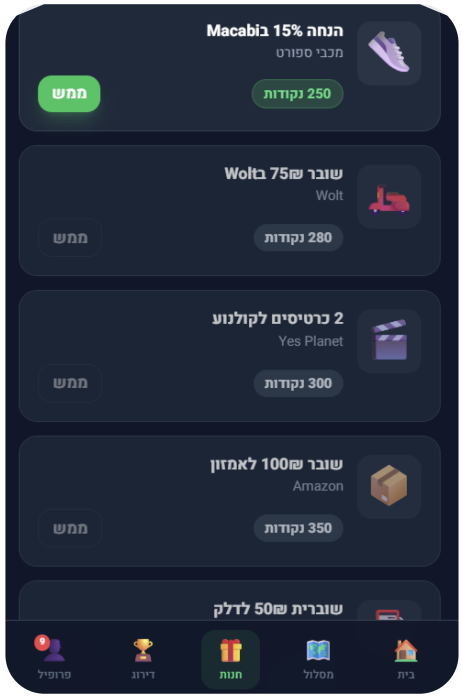


Ranking -> Member Leaders Table:

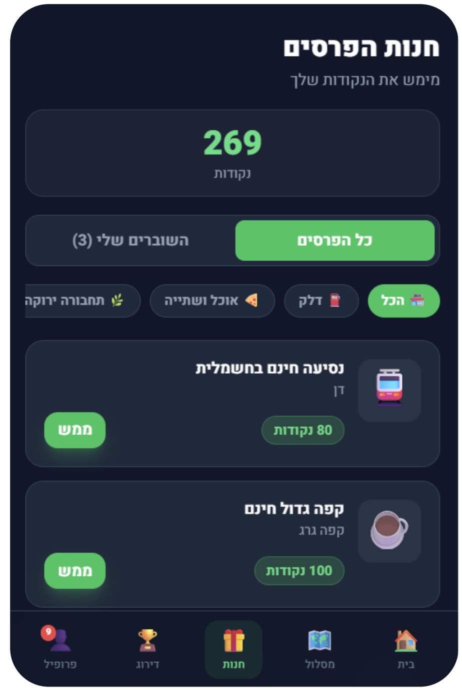
Ranking-> National Leaderboard:



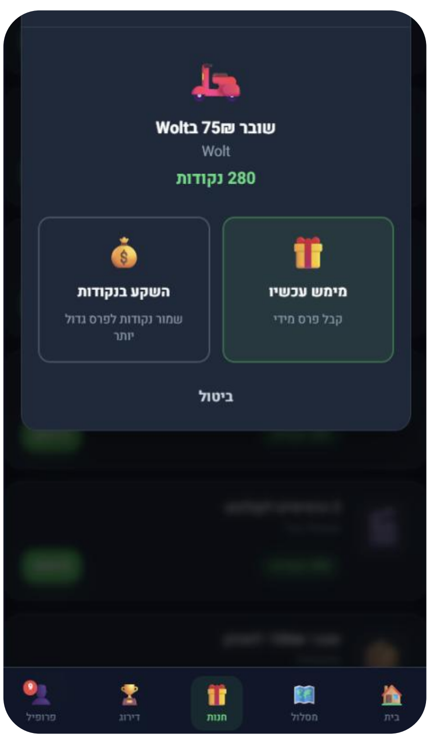
The Prize Shop - Continued:



The Prize Shop:

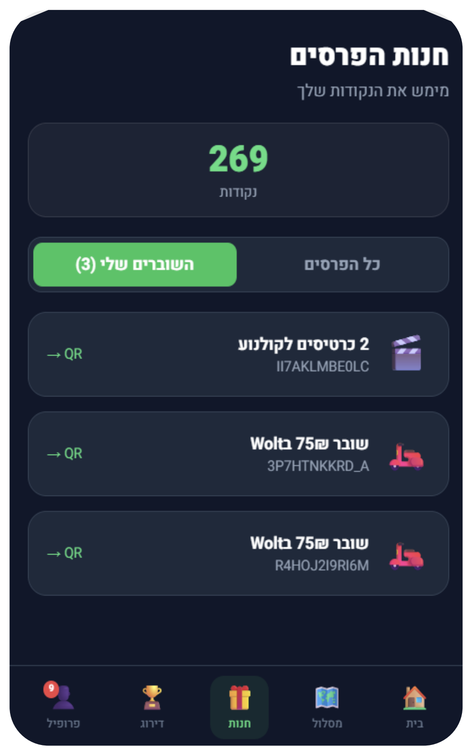


Shop > buy a voucher:

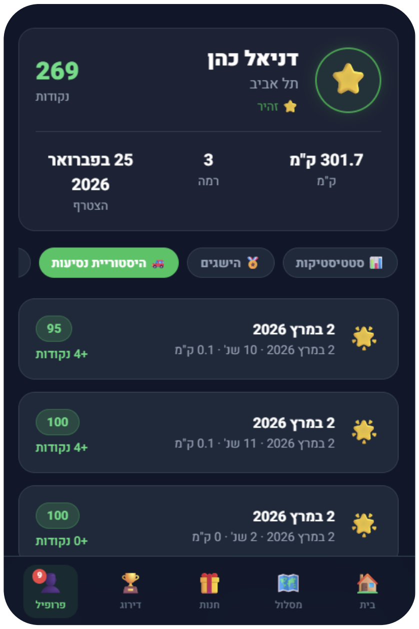


Shop > Presentation of a business voucher:

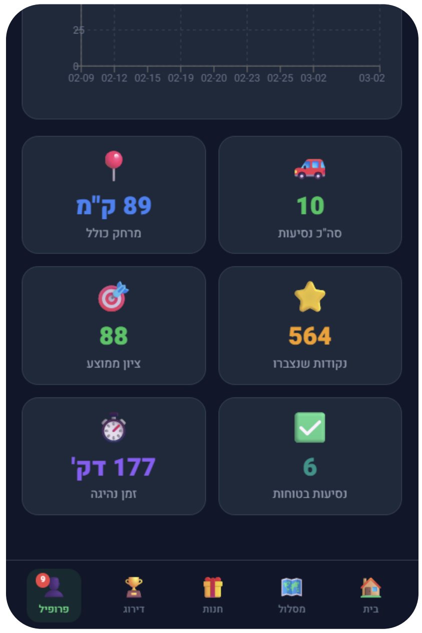
Shop > my vouchers:



Travel Profile:

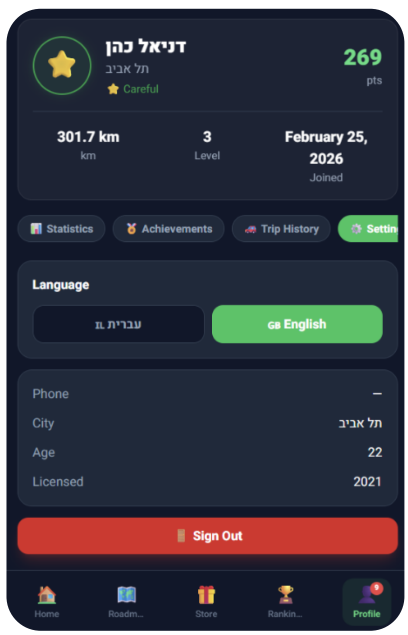


Follow-up:

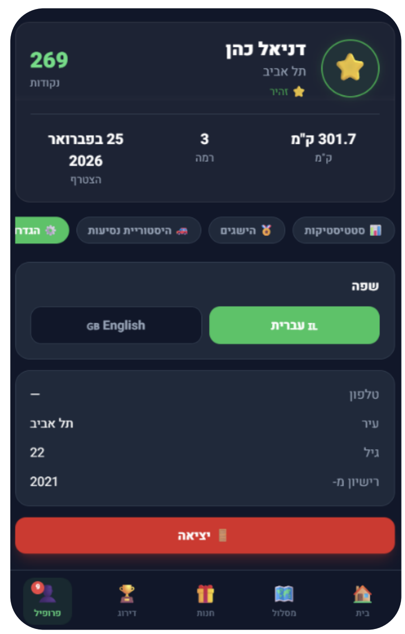


Profile:

Profile -> Definitions -> EN:



Profile -> Settings -> HE:



Appendix B - Glossary

IMU	Inertial Measurement Unit. Inertial unit of measurement (Accelerometer + Gyroscope).
OTP	One-Time Password. One-time password.
CARMA Score	Specify a single trip.
CARMA Points	Accumulating points that can be redeemed.
BT	Bluetooth. Short-range wireless communication protocol.
QR Code	Quick Response Code. 2D code for scanning.
Foreground Service	Background service with user indication.
Local Cache	Local conversion on the device.
Payload	A data packet is sent to the server.
GDPR	General Data Protection Regulation. European Privacy Protection Regulations.

Appendix C – CARMA Score Algorithm

1. Introduction

The CARMA Score algorithm calculates a score for each trip made by the user. The score reflects the level of safety of driving and serves as a basis for calculating CARMA points that can be redeemed in the store.

Each trip starts with a perfect score of 100. Points are deducted according to exceptional events detected by the IMU and GPS sensors, taking into account the severity of the incident and the duration of the trip. Safe driving during risk hours is rewarded with bonus points.

2. Prerequisites for calculating a score – (Appendix E)

3. Types of events

The system identifies two types of behavior, which differ in the way the punishment is calculated:

- **Type:** Discrete (Momentary)
  - **Events:** Heavy braking, abnormal acceleration, sharp turning, lane departure.
  - **Calculation Method:** Event Count Multiplied Penalty Multiplied Severity.
  - **Explanation:** The event takes place and ends in an instant.
- **Type:** Ongoing (time-based)
  - **Events:** Speed exceedance, phone use.
  - **Method of calculation:** Ratio of behavior time to total travel.
  - **Explanation:** The behavior lasts for a long time.

4. Noise Filter

GPS and IMU data may contain measurement errors, especially in tunnels, near tall buildings, or when the satellite signal is weak. To prevent false identifications, the system implements the following rules:

- **Rule:** Minimum time threshold for an ongoing event

  Description: A velocity exceeding the limitwill only be counted ifat least 3 consecutive seconds are taken. A single sample exceeding the threshold will be discarded.

- **Rule:** Minimum time threshold for a discrete event

  Description: Braking, acceleration, and cornering will only be counted if the sensor value exceeds the threshold for at least 0.5 consecutive seconds (5 samples at 10Hz).

- **Included:** GPS Accuracy Filtering

  Description: A GPS sample with an accuracy greater than 20 meters will be marked as unreliable. Speed and deviation events will not be counted based on unreliable samples.

- **Rule:** Smoothing

  Description: IMU sensor values will pass a Moving Average over a window of 5 samples (0.5 seconds) before an exceeding threshold estimate.

5. Deduplication rule

Sharp turn (EVT_TURN) and lane departure (EVT_SWERVE) events may be detected simultaneously on the same physical maneuver because they are measured by two different sensors.

- **Rule:** If EVT_TURN and EVT_SWERVE are detected within a 2-second window of each other, only the event with the highest penalty (the greatest value of P multiplied by W) will be counted. The second event will be deleted from the list of events.

6. The Total Reduction Formula

The total reduction consists of three components:

Total_Penalty = [Σ(P × W) × N] + [(T_phone / T_trip) × 40 × M] + [(T_speed / T_trip) × 35 × W_avg]

Legend of Variables:

Basic punishment for each isolated incident	**P**
Hardness coefficient according to the intensity of the exception	**W**
Promotes normalization for travel duration	**N**
Total seconds of phone usage	**T_phone**
Speed Exceeded Total Seconds	**T_speed**
Total travel time in seconds	**T_trip**
Frequency Multiplier Phone Lifts	**M**
Weighted severity average of speed anomalies	**W_avg**

When:

- The first element relates to discrete events (braking, accelerating, turning, diversion).
- The second component relates to the use of the telephone.
- Thethird component refers to the speed exceedance.

Details of each component appear in sub-Appendix C1.

7. Calculating a trip score (CARMA Score)

CARMA_Score = clamp(round(100 - Total_Penalty), 0, 100)

The score is rounded to an integer so that the rank accurately reflects performance.

Grades:

- **90 and above:** Excellent.
- **75 to 89:** Good.
- **55 to 74:** Passable.
- **Below 55:** Needs improvement.

8. Calculation of Points to Redeem (CARMA Points)

CARMA_Points = floor(Score × F_dist × R_risk × B_level)

When:

- **Score –** Ride score, 0 to 100 (Section 7)
- **F_dist –** Logarithmic distance coefficient, decreasing return (section 9)
- **R_risk –** Risk multiplier, bonus weighted according to travel hours (section 10)
- **B_level –** Boost Rank, Multiplier for Advanced Drivers (Appendix D)

The dots are rounded downwards (floor). The reason: pips are an internal currency that can be exercised, and rounding down prevents inflation.

9. Distance coefficient

F_dist = min(ln(D_km + 1) / ln(11), 2.0),

When D_km is the actual driving distance in kilometers.

The logarithmic curve rewards longer trips, but at a decreasing rate. A 2.0 ceiling prevents extreme values.

Example table of values:

Promoter	Distance
0.29	1 km
0.58	3 km
0.75	3.1 miles
1.00	10 km
1.27	20 km
1.64	31.1 miles
2.00 (Ceiling)	100 km and above

Note: The measured distance is the actual driving distance (not linear distance). A minimum linear distance (500 m) is only tested under the prerequisites.

10. Weighted risk multiplier

On trips that cross different time windows, the multiplier is calculated as a time-weighted average:

R_risk = (T_weekend_night × 2.0 + T_regular_night × 1.5 + T_day × 1.0) / T_total

When:

- T_weekend_night – Minutes of Driving on a Weekend Night
- T_regular_night – Minutes of Driving on a Normal Night
- T_day – Minutes of daytime driving
- T_total – Total minutes drive

Time windows:

- Weekend night (Thursday and Friday 23:00 to 04:00): Multiplier 2.0.
- Regular night (any other night 23:00 to 04:00): Multiplier of 1.5.
- Daylight hours (any other hour): Multiplier of 1.0.

Rationale: According to data from the Central Bureau of Statistics and Green Light, the night hours, and especially on weekend nights, are the most dangerous on Israel's roads. A driver who drives carefully during these hours is rewarded with bonus points.

Example: A ride that starts on Thursday at 22:40 and ends at 23:10 (30 minutes):

22:40 to 23:00 – 20 minutes, multiplier 1.0 (day)

23:00 to 23:10 – 10 minutes, multiplier 2.0 (Thursday night)

R_risk = (20 × 1.0 + 10 × 2.0) / 30 = 40 / 30 = 1.33 (risk multiplier)

Sub-Appendix C1 – Event Tables, Sensor Thresholds and Parameters

I. Discrete Incidents – Penal Table and Identification Thresholds

Basic punishment	Minimum time threshold	Detection Threshold	Sensor	Event Name
5	0.5 Monday	Above $m/s^2$4.5	Accelerometer Y-axis	EVT_BRAKE Strong braking
3	0.5 Monday	Above $m/s^2$4.0	Accelerometer Y-axis	EVT_ACCEL Unusual acceleration
3	0.5 Monday	Above $m/s^2$3.5	Accelerometer X-axis	EVT_TURN A sharp turn
5	3 sec	Over 15 degrees per second	GPS Heading	EVT_SWERVE Sudden lane deviation

Notes:

- IMU values are smoothed (Moving Average over 5 samples) before an exception is evaluated.
- EVT_SWERVE requires 3 consecutive seconds (GPS noise filtering).
- EVT_TURN and EVT_SWERVE events are subject to the rule of preventing double counting (Appendix C, Section 5).

II. Promotes hardware for discrete events

For each discrete event, a multiplier is determined according to the intensity of the exceeding threshold:

- **Light Level:** 0% to 30% exceeding the threshold. Multiplier: 1.0.
- **Medium Level:** 30% to 70% exceeding the threshold. Multiplier: 1.5.
- **Severe level:** Over 70% exceeding the threshold. Multiplier: 2.0.

Example (brake threshold = $m/s^2$4.5):

m $/s^2$ value 5.0, exceeding 11%, light level, W multiplier = 1.0

6.5 m/s^2 value, 44% exceeded, medium level, W multiplier = 1.5

8.0 m/s^2 value, exceeding 78%, severe level, W multiplier = 2.0

III. Normalization of travel duration – applies to discrete events only

N = clamp(15 / T_minutes, 0.3, 3.0)

When T_minutes is the duration of the trip in minutes.

- **Duration:** 3.0 (Ceiling) – Increased penalty for a short drive.
- **Duration 5 minutes:** 3.0 (ceiling) – increased penalty.
- **10 min Duration**: 1.5
- **Duration 15 minutes:** N = 1.0 : attribution, unchanged.
- **Duration 30 minutes:** N = 0.5 : Reduced penalty.
- **Duration of 60 minutes or more:** 0.3 (floor) – minimum penalty.

This normalization does not apply to ongoing events (telephone and speed) – they are inherently normalized by the ratio of event time to total travel time.

IV. Using the phone – a detailed calculation

P_phone = (T_phone / T_trip) × 40 × M_pickup

M_pickup = min(1 + 0.1 × max(0, N_pickups - 1), 2.0)

When:

- T_phone – The total number of seconds that the device is raised while driving
- T_trip – Total travel time in seconds
- 40 – Promotes Basic Punishment
- N_pickups – Number of times the device was raised (separate events)
- M_pickup – Frequency multiplier (ceiling: 2.0)

A condition for identifying "lifting a device" – the following two conditions must be met at the same time:

- **Angle change:** Gyroscope pitch above 30 degrees for at least 2 consecutive seconds.
- **Screen interaction:** Touch Events or screen unlock—at least one event is detected.

Explanation: Pitch change only causes false identifications on steep roads, bridges, and interchanges. Integration with screen interaction ensures that the driver has indeed used the phone.

Phone on Mount: When the phone is mounted, a pitch change will not be considered a level. The penalty will only be calculated based on active screen time, with a M_pickup of 1.0.

Examples (60 seconds of use from a 25-minute drive):

- **1 long level:** 1 level, M=1.0, Penalty=1.6.
- **3 20-second levels:** 3 levels, M=1.2, Penalty=1.92.
- **6 10-second levels:** 6 levels, M=1.5, Penalty=2.4.
- **The 11 levels and above:** M=2.0 (ceiling), penalty=3.2.

V. Speed Exceeding – Detailed Calculation

P_speed = (T_speed / T_trip) × 35 × W_avg,

When:

- T_speed – Total seconds in speed exceeding the threshold (above the threshold)
- T_trip – Total travel time in seconds
- 35 – Promotes Basic Punishment
- W_avg – Time-Weighted Average Severity

Detection threshold: GPS speed above the speed limit of the road plus 10 km/h.

Allowable speed data source: An external interface for allowable speed data (such as the Google Roads API or OpenStreetMap).

Backup rule: If there is no speed limit data available for the current road, a speed limit will not be counted for that segment. No penalties without reliable information.

Noise Filtering: A speed deviation will only be counted if it lasted at least 3 consecutive seconds, and all GPS samples during that time exceeded the accuracy threshold (less than 20 meters).

Severity levels for speed exceeding the threshold (in kilometers per hour above the threshold):

- Exceeding 1 to 15 km/h: Slight level, W = 1.0. **Example (90 allowed, 100 threshold):** 101 to 115 km/h.
- Exceeding 15 to 30 km/h: Medium level, W = 1.5. **Example:** 116 to 130 km/h.
- Exceeding more than 30 km/h: Severe level, W = 2.0. **Example:** 131 km/h or above.

Weighted average severity – if the driver has exceeded different levels during the trip:

W_avg = Σ(W_segment × T_segment) / T_speed_total

When W_segment is the severity level of each segment, T_segment it lasted in seconds, and T_speed_total is the total of abnormal seconds.

VI. Customizable parameter table via admin interface

Discrete Events:

- **Braking threshold:** Default m/s²4.5, range 3.0–6.0.
- **Acceleration threshold:** Default m/s²4.0, range 2.5–5.5.
- **Turnout threshold:** Default m/s²3.5, range 2.0–5.0.
- **Deviation threshold:** default sec/15°, range 10–25.
- **Dedup window : Default** 2 seconds, range 1–5.

Ongoing Events:

- **Phone lift threshold (duration):** Default 2 seconds, range 1–5.
- **Phone lift threshold (angle):** Default 30°, range 20–45.
- **Phone penalty coefficient:** Default 40, range 20–60.
- **Level multiplier ceiling:** Default 2.0, range 1.5–3.0.
- **Minimum speed exceedance:** default +10 km/h, range +5 to +20.
- **Speed penalty coefficient:** Default 35, range 20–50.

Noise Filtering:

- **Minimum time for discrete event:** Default 0.5 sec, range 0.3–1.0.
- **Minimum time for an ongoing event:** Default 3 seconds, range 2–5.
- **Maximum GPS accuracy :** Default 20m, range 10–50.
- **IMU smoothing window:** Default 5 samples, range 3–10.

Normalization and Promoters:

- **Reference drive (T_base):** Default 15 minutes, range 10–30.
- **Ceiling N_dur:** Default 3.0, range 2.0–5.0.
- **Floor N_dur:** Default 0.3, Range 0.1–0.5.
- **Ceiling F_dist:** Default 2.0, range 1.5–3.0.
- **Standard night multiplier:** Default 1.5, range 1.0–2.0.
- **Weekend Night Multiplier:** Default 2.0, range 1.5–3.0.

Threshold and Protection Conditions:

- **Minimum travel duration:** Default 2 minutes, range 1-5.
- **Minimum linear distance:** Default 500 m, range 200–1000.
- **Rewarded trips per day:** Default 6, range 3–10.

Sub-Appendix C2 – A Complete Example of Calculating a Travel Score

Travel Details

- **Day and time:** Thursday, 22:40 to 23:10.
- **Duration:** 30 minutes (1,800 seconds).
- **distance**: 14 km.
- **Linear distance from the starting point:** 9.5 km.
- **Trip number per day:** 3.
- **Speed limit on the road:** 90 km/h (threshold = 100 km/h).
- **Driver Level 5 – :** Sharp (Boost Rank = 1.0).

Checking Threshold Conditions

I. **Recognition as a ride:** The driver drove over 10 km/h for 30+ consecutive seconds – the ride was detected.

II. **Prerequisites:**

   a. **Travel time:** 30 minutes (over 2 minutes) – passing.

   b. **Linear distance:** 9.5 km (over 500 meters) – passing.

III. **Daily quota: Trip** number 3 out of 6 – passing.

The ride meets all the conditions, you will receive a score and earn points.

Step 1 – Identify Events

- **Discrete events (before filtering):**

  i. **Event 1:** Strong braking.

  Measured value: 5.2 m/s^2. Threshold: 4.5. Deviation: 15%. Level: Easy. P=5. W=1.0. Time stamp: 22:47:12.

  Noise Filtering: An exceeding value lasts 0.8 seconds (over 0.5 seconds) – is filtered.

  ii. **Event 2:** A sharp turn.

  Measured value: 5.0 m/s^2. Threshold: 3.5. Deviation: 43%. Level: Medium. P=3. W=1.5. Time stamp: 22:53:30.

  iii. **Event 3:** Lane deviation.

  Measured value: 18°/s. Threshold: 15. Exceeded: 20%. Level: Easy. P=5. W=1.0. Time stamp: 22:53:31.

  Noise Filtering: Lasts 4 consecutive seconds (over 3 seconds), GPS accuracy – 8 meters (under 20) – is filtered.

- **:D Edup Testing**

  Event 2 (turn, 22:53:30) and Event 3 (deviation, 22:53:31) are identified in a window of 1 second (less than 2 seconds).

  - **Penalties for penalties:** 3 × 1.5 = 4.5.
  - **Deviation penalty:** 5 × 1.0 = 5.0.

  The deviation (5.0) is higher than the redirect (4.5), so the redirect is deleted.

- **Final discrete events (after filtering and dedup-:**
  - **Event 1:** Strong braking. P=5. W=1.0.
  - **Event 3:** Lane deviation. P=5. W=1.0.
- **Ongoing Events:**
  - **Phone usage:** 50 seconds total, 4 levels (pitch over 30° combined with touch screen).
- **Speed Exceeded:**
  - **Segment A:** 108 km/h for 60 seconds (8 km/h exceeding the threshold, light levelW=1.0,)
  - **Segment B:** 118 km/h for 20 seconds) 18 km/h, medium level (W = 1.5,

  Note: The two speed deviation segments lasted for over 3 consecutive seconds and the GPS accuracy was normal – they were noise filtered.

Step 2 – Calculating the penalty for discrete events

- **Normalization of travel duration:**

  N = 15 / 30 = 0.5

  - **Event 1 (Braking):** 5 × 1.0 × 0.5 = 2.5
  - **Event 3 (Imp Vol):** 5 × 1.0 × 0.5 = 2.5

  Total discrete events: 5.0

Step 3 – Calculate the penalty for using a phone

- **Level Multiplier:** M = min(1 + 0.1 × max(0, 4 - 1), 2.0) = min(1.3, 2.0) = 1.3
- **Phone Penalty:** P_phone = (50 / 1,800) × 40 × 1.3 = 0.0278 × 40 × 1.3 = 1.44

Step 4 – Calculating a Speed Violation Penalty

- **Weighted severity average:** W_avg = ((1.0 × 60) + (1.5 × 20)) / 80 = 90 / 80 = 1.125
- **Speed penalty:** P_speed = (80 / 1,800) × 35 × 1.125 = 0.0444 × 35 × 1.125 = 1.75

Step 5 – Total penalty and travel score

- **Total Reduction**: 5.0 + 1.44 + 1.75 = 8.19
- **Score**: clamp(round(100 - 8.19), 0, 100) = round(91.81) = 92
- **Rank**: Excellent.

Step 6 – Calculating CARMA Points

- **Distance coefficient:** F = min(ln(14 + 1) / ln(11), 2.0) = min(1.13, 2.0) = 1.13
- **Weighted risk multiplier (cross-window driving):**

  22:40 to 23:00 – 20 minutes, multiplier 1.0 (day)

  23: 00 to 23:10 – 10 minutes, multiplier 2.0 (Thursday night)

  R_risk = (20 × 1.0 + 10 × 2.0) / 30 = 40 / 30 = 1.33
- **Boost Rank**: Level 5 Driver = 1.0
- **Final result:**

  Points = floor(92 × 1.13 × 1.33 × 1.0) = floor(138.25) = <u>138 CARMA Points</u>

Trip summary – as presented to the user

- **Travel score:** 92 (Excellent).
- **Points earned:** +138 CARMA Points .
- **Duration:** 30 minutes.
- **distance:** 14 km.
- **Bonus:** ×1.33 (part of the trip during risk hours).

Personal insights to be presented at the end

- One hard braking – "Keep a safe distance from the vehicle in front of you."
- Lane deviation – "Stay focused on the lane."
- Use your phone for 50 seconds on all 4 levels – "Put your phone down while driving."
- Exceeding the speed limit of 80 seconds – "Observe the speed limit (90 km/h)."
- "Bonus × 1.33 for safe driving in the evening and at night – well done!"

Comparing scenarios – the same ride under different conditions

- **As it happened:** score 92, distance coefficient 1.13, risk multiplier 1.33, boost 1.0, total 138 points.
- **The entire ride after 23:00 (Thursday night):** score 92, coefficient 1.13, multiplier 2.0, boost 1.0, total 207 points.
- **The entire ride on a normal day:** a score of 92, a coefficient of 1.13, a multiplier of 1.0, a boost of 1.0, a total of 103 points.
- **Perfect ride (same conditions):** score 100, coefficient 1.13, multiplier 1.33, boost 1.0, total 150 points.
- **Level 9 driver (same drive):** 92 score, coefficient 1.13, multiplier 1.33, boost 1.1, total 152 points.
- **Level 10 driver (same drive):** Score 92, coefficient 1.13, multiplier 1.33, boost 1.2, total 165 points.
- **No Dedup :** 90 Score, Distance Coefficient 1.13, Risk Multiplier 1.33, Boost 1.0, Total 135 Points.

The driver lost 12 points (150 minus 138) due to 4 abnormal behaviors, about 8% less than a perfect ride.

The Dedup rule saved the driver 3 points that would have been unjustly deducted.

Appendix D – Transition Table Between Levels

- **User Rank System - CARMA Roadmap**

  Principles of Route Design The grading system is based on a moderate exponential progression curve. The goal is to enable a quick transition between the first stages to encourage new user engagement, and to make the transition more challenging and rewarding in the advanced stages.

- **Daily Point Rate Estimate**
  - Occasional driver (one short trip): about 60 to 80 points per day.
  - Daily driver (2-3 trips): about 150 to 250 points per day.
  - Heavy driver (many or long trips): about 300 to 500 points per day.

- **Progress Stages Table**

  The system manages a track of 10 user levels, with the transition from one level to another conditional on the accumulation of points:

Estimated time to the next step	Points for the next step	Accumulating Points	Stage Name	Level
~3 days	500	0	Begins	1
~Week	1,000	500	Careful	2
~2 weeks	2,000	1,500	Concentrated	3
~3 weeks	3,500	3,500	Skilled	4
~5 weeks	5,000	7,000	sharp	5
~7 weeks	8,000	12,000	Expert	6
~2.5 months	12,000	20,000	Wizard	7
~3.5 months	18,000	32,000	Master	8
~5 months	25,000	50,000	General of the Road	9
-	-	75,000	Legend	10

Benefits and features that open according to the step of progression, providing added value to the driver and making new capabilities available in the app:

Unfolding Features	Basic Store Discount	Level
Basic access to the store	-	1-2
Community comparison	5%	3
Extended Travel Statistics	5%	4
Exclusive Benefits	10%	5
"Sharp Driver" badge on your profile	10%	6
Early access to new benefits	15%	7
Exclusive Premium Benefits	15%	8
"Road General" badge + Boost points x 1.1	20%	9
"Legend" Badge + Points Boost X 1.2	25%	10

- **Retention rules and stage management**
  - **Rank Retention**: The rank is based on the total points accumulated over time. Redeeming points on Marketplace – does not lower the user's rank.
  - **Streak bonus:** A driver who maintains 7 consecutive days of driving with an average CARMA score of 75 or higher will win a one-time bonus of 200 points.
  - **"Legend" level management:** After reaching level 10, the user continues to accumulate points for advancing in the national leaderboard.

Appendix E – Prerequisites for calculating a score

A trip will only receive a score and earn points if it meets three types of conditions: Ride Recognition, Travel Quality, and Usage Restriction.

I. **Recognition as Travel**

   This condition determines when the system starts and ends detecting a trip, in accordance with the definition in clause 4.4.5 of the specification document:

   Condition: Continuous movement over 10 km/h for at least 30 seconds.

   Explanation: Only then does the system begin to sample and analyze the ride. Short stops (traffic light, traffic jam) do not stop the ride – the ride ends when the speed is less than 10 km/h for 3 minutes straight.

II. **Prerequisites for calculating grades and points**

   Even after the system detects a trip, it will calculate a score and award points only if the trip meets the following two conditions:

   i. **Travel duration** – at least 2 minutes from the moment the identification begins until the end of the trip.

      Explanation: Shorter trips (e.g., reparking, entering a gas station) do not represent significant driving behavior.

   ii. **Linear distance** – at least 500 meters in a straight line between the starting point and the end point.

      Explanation: Preventing the use of the system by driving in circles. This is a linear (aerial) distance, not the actual driving distance.

III. **Daily Usage Limit**

   Condition: Maximum of 6 rewarded trips on a calendar day.

   Explanation: Preventing multiple artificial trips. The seventh trip or later will go down in history and receive a score, but will not earn CARMA Points.

Summary – What happens with travel that does not meet the conditions:

- A trip that did not meet condition A will not be recorded at all.
- A trip that did not meet condition B will be recorded in the travel history, but will not receive a score and will not earn points.
- A trip that exceeds condition C will be registered and will receive a score, but will not earn points.